MetaMining

A Pure-Mojo Latent Space Explorer

Stop chatting with LLMs.
Start mining their topology.

Gemini Notebook

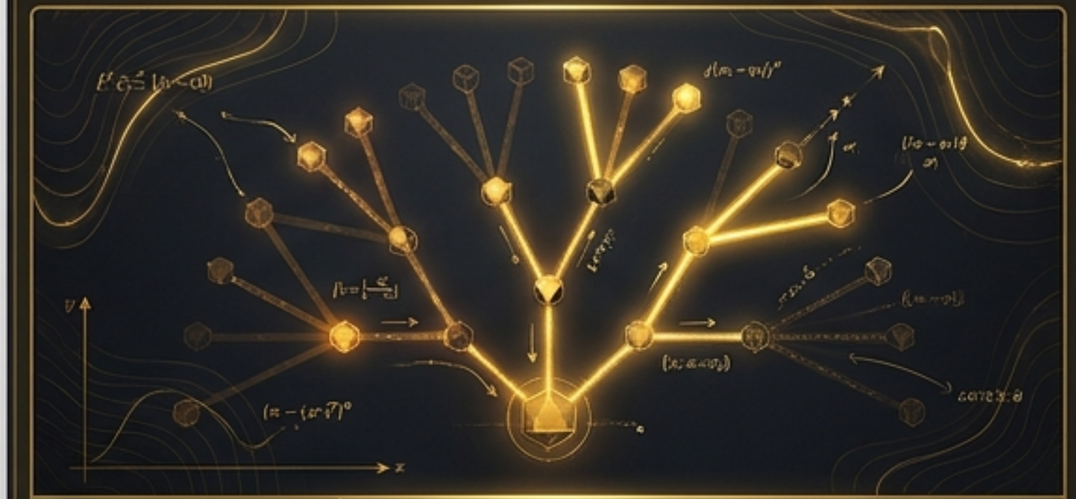
The Paradigm Shift

Traditional Chat



Structure:	Single-thread, linear progression.
Dependency:	Highly prompt-dependent.
Output:	Flat text generation.
Waste:	Dead-ends and suboptimal branches are discarded.

MetaMining Cartography

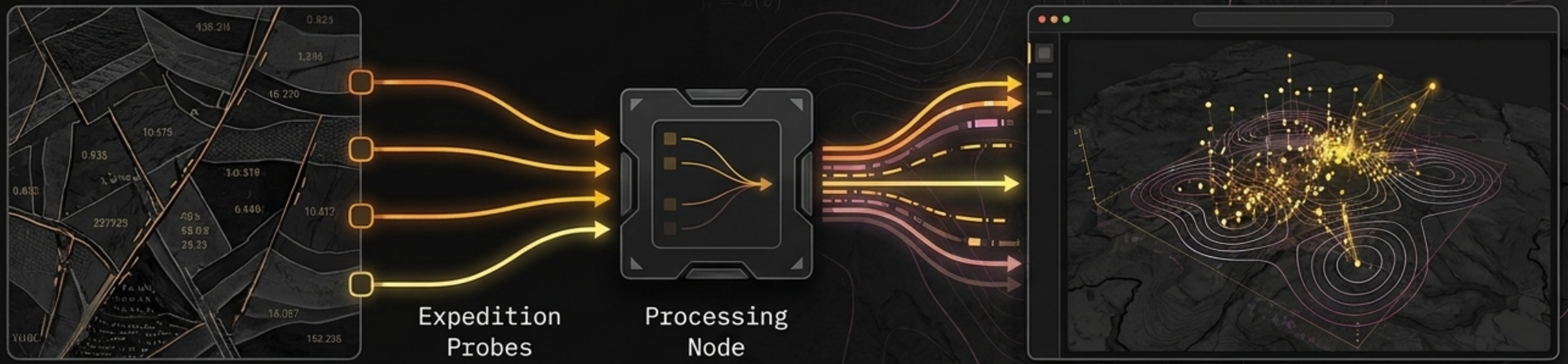


Structure:	Multi-branch Hive (Beam Search up to 8 candidates).
Dependency:	Topology-steered spatial exploration.
Output:	Mathematical Cartography and 3D surface maps.
Retention:	Immortalized **Gold Veins** stored on the **Pareto frontier**.

Gemini Notebook

The MetaMining Expedition

Claim Map Dashboard



01	Deploy the Hive	02	Score the Strike	03	Map the Dig
Launch independent expedition branches equipped with persona lenses and seed chains.		Evaluate every finding against a strict mathematical blend of novelty × coherence.		Plot coordinates live on a 3D topological claim map, rendering the shape of the LLM's knowledge.	

Gemini Notebook

The Engine Architecture

Dual Environment Deployment

Operates live via Meta Responses API (/v1/responses, /v1/embeddings) or offline using a deterministic Sim Backend.

Native Engine Core

HTTP server, JSON arena parser, TLS 1.3 client, PCA, Topology, and Vector Steering algorithms.

Zero Bloat Foundation

Fork/exec-free process control. Interfaces solely with libc and the system OpenSSL via Mojo FFI.

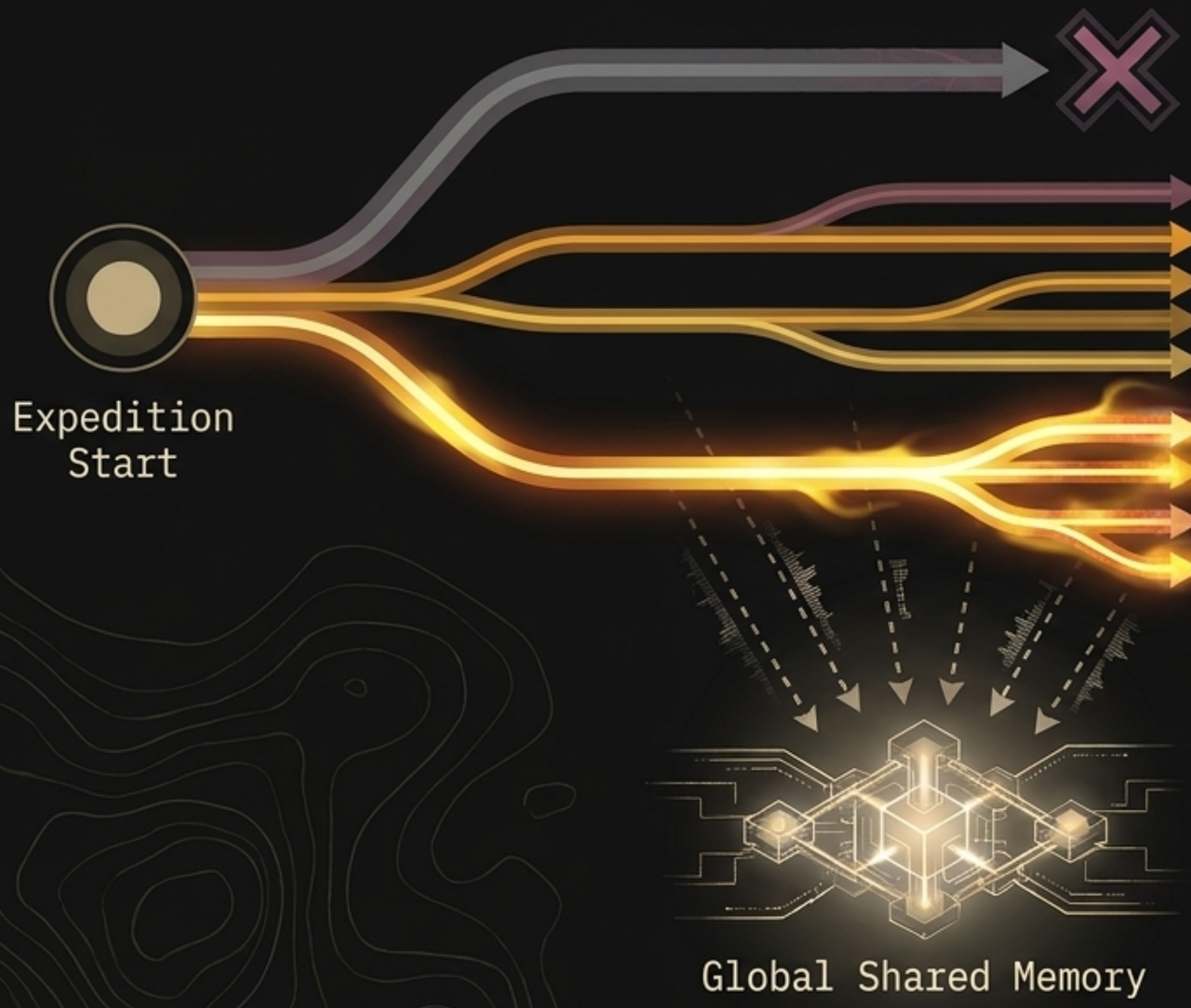


100% Pure Mojo

Designed for extreme speed and lightweight execution. No Python dependencies.

Gemini Notebook

The Hive 10x



Branches sync every 4th step. Novelty is scored globally against the whole hive, preventing redundant digging.

Population Control

- Embedded detectors kill threads stuck on plateaus or incoherence.
- Vein-miner branches are dynamically spawned at gold strikes.
- Workers cool down to lock onto discovered veins.

Gemini Notebook

Scoring the Strike

```
Yield = KDE Novelty (τ=0.20, γ=0.97)
      + Lexical Entropy
      + Vector Steering
      + Topological Blend

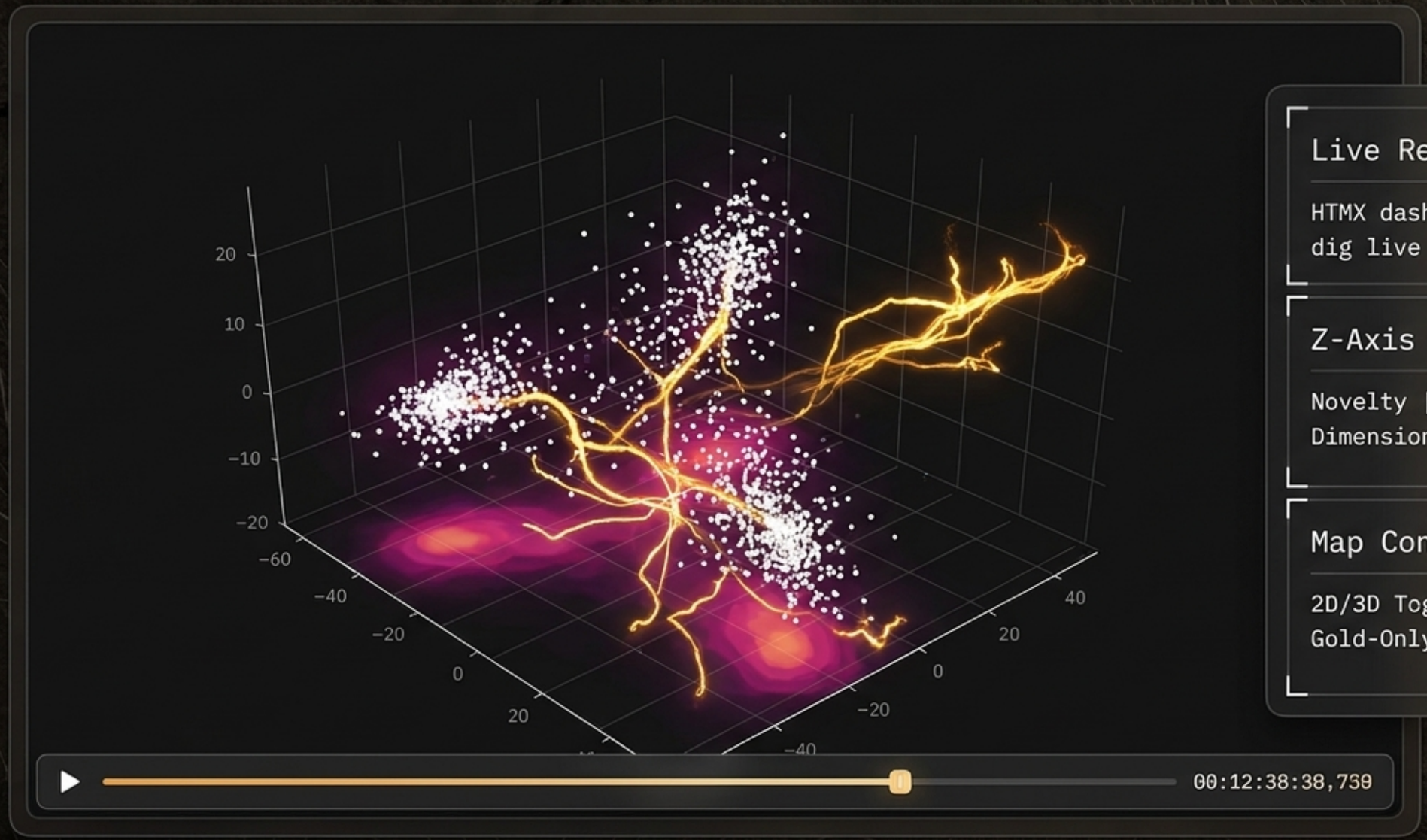
Coherence Rating: LLM Judge votes scored [1,10]
```



Non-dominated claims are automatically saved directly to the mathematical Pareto frontier.

Gemini Notebook

Cartography: The Claim Map



Live Rendering

HTMX dashboard plotting the dig live via SSE fan-out.

Z-Axis Metric Toggle

Novelty | PCA-Z | Cohomological Dimension | Coherence | Density

Map Controls

2D/3D Toggle | Branch Filters | Gold-Only Mode

Gemini Notebook

The Shape of Knowledge

Raw Data → Clusters



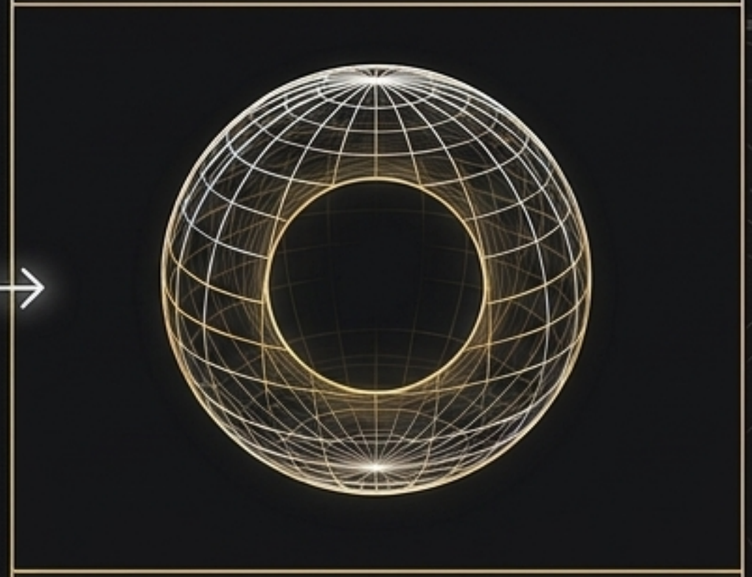
Vietoris-Rips topology groups points into Merge Trees (novelty peaks and basins).

Clusters → Loops (Betti 1)



Data forms repetition rings indicating cyclical knowledge structures.

Loops → Voids (Betti 2)



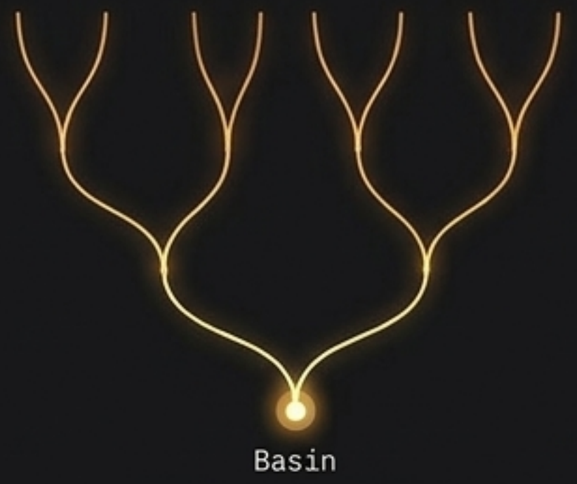
Reveals enclosed, unexplored 3D chambers within the latent space.

MetaMining calculates exact Vietoris-Rips topology to find basins of insight, repetition rings, and unexplored knowledge voids.

Gemini Notebook

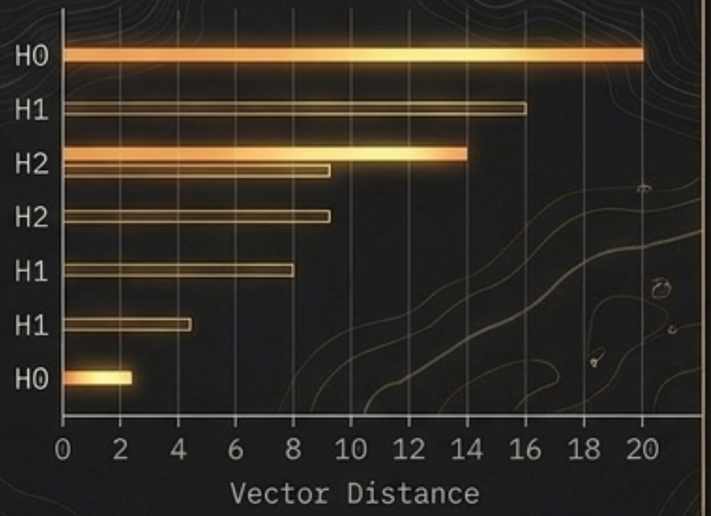
The Shape Laboratory: Diagnostics

Merge Tree



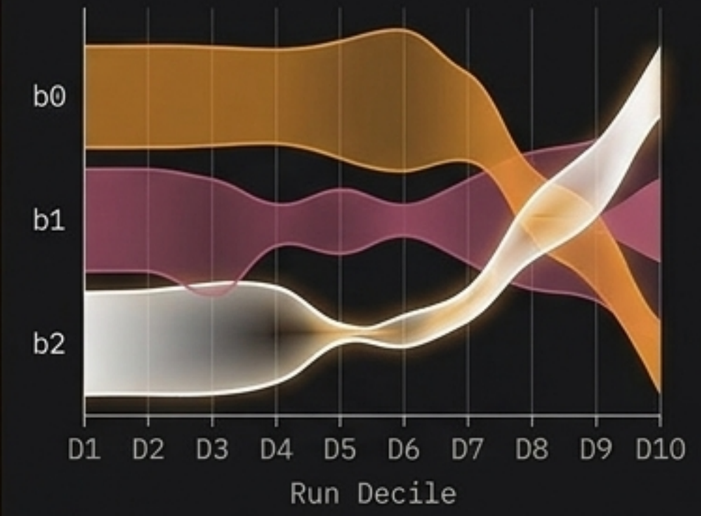
Visualizes novelty peaks joining downward. Basins indicate which hill each data point drains to.

Barcode + Betti



Persistence bars and Betti curves over vector distance. Bold bars indicate 8-fold resampling confidence.

Run Timeline

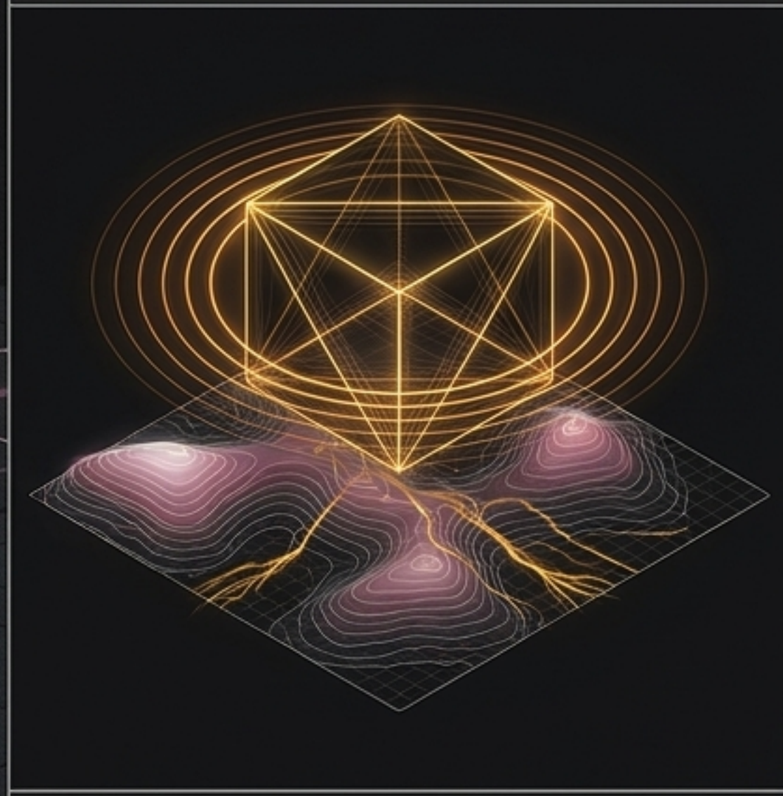


Ribbon charts tracking Betti numbers over time, showing loops forming versus voids closing.

Gemini Notebook

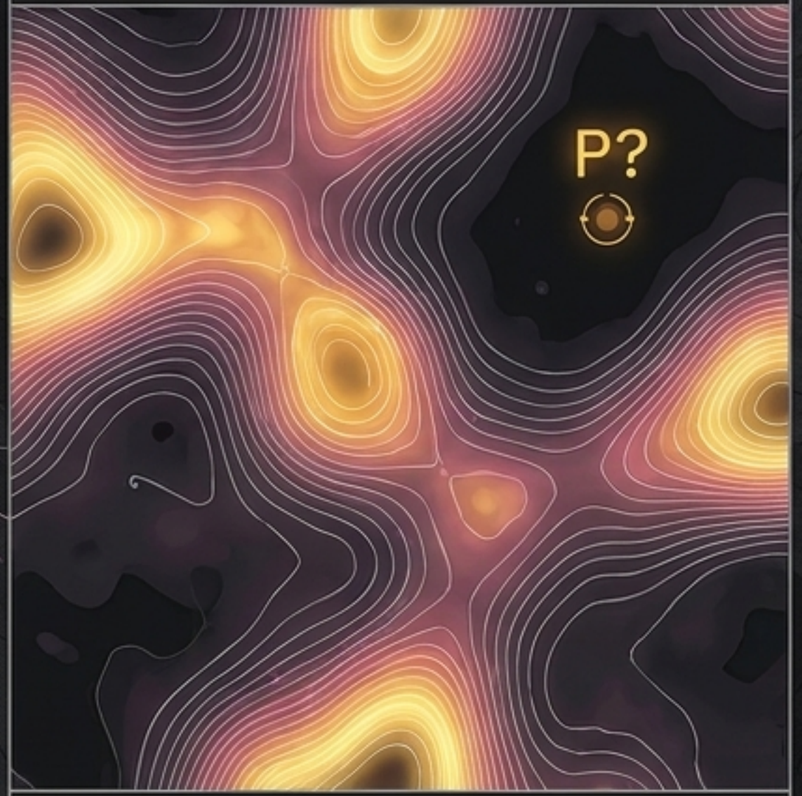
The Shape Laboratory: Discovery

Loops + Void



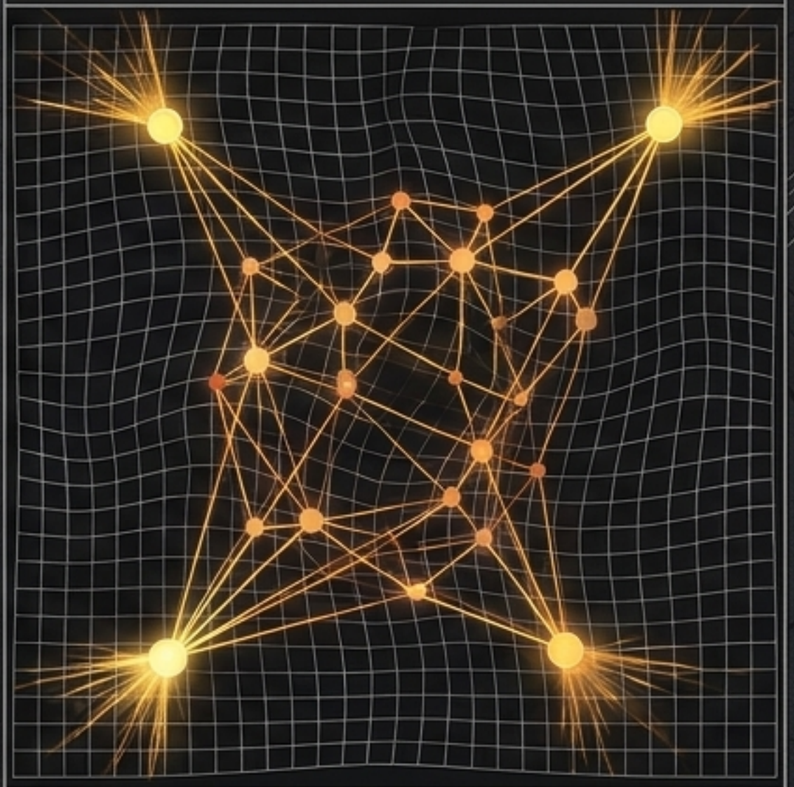
Maps repetition rings and the unexplored void chamber at median scale over the terrain.

Density + Contours



KDE heat maps and iso-contours (bandwidth 5). Highlights sparse gaps marked for automated probing.

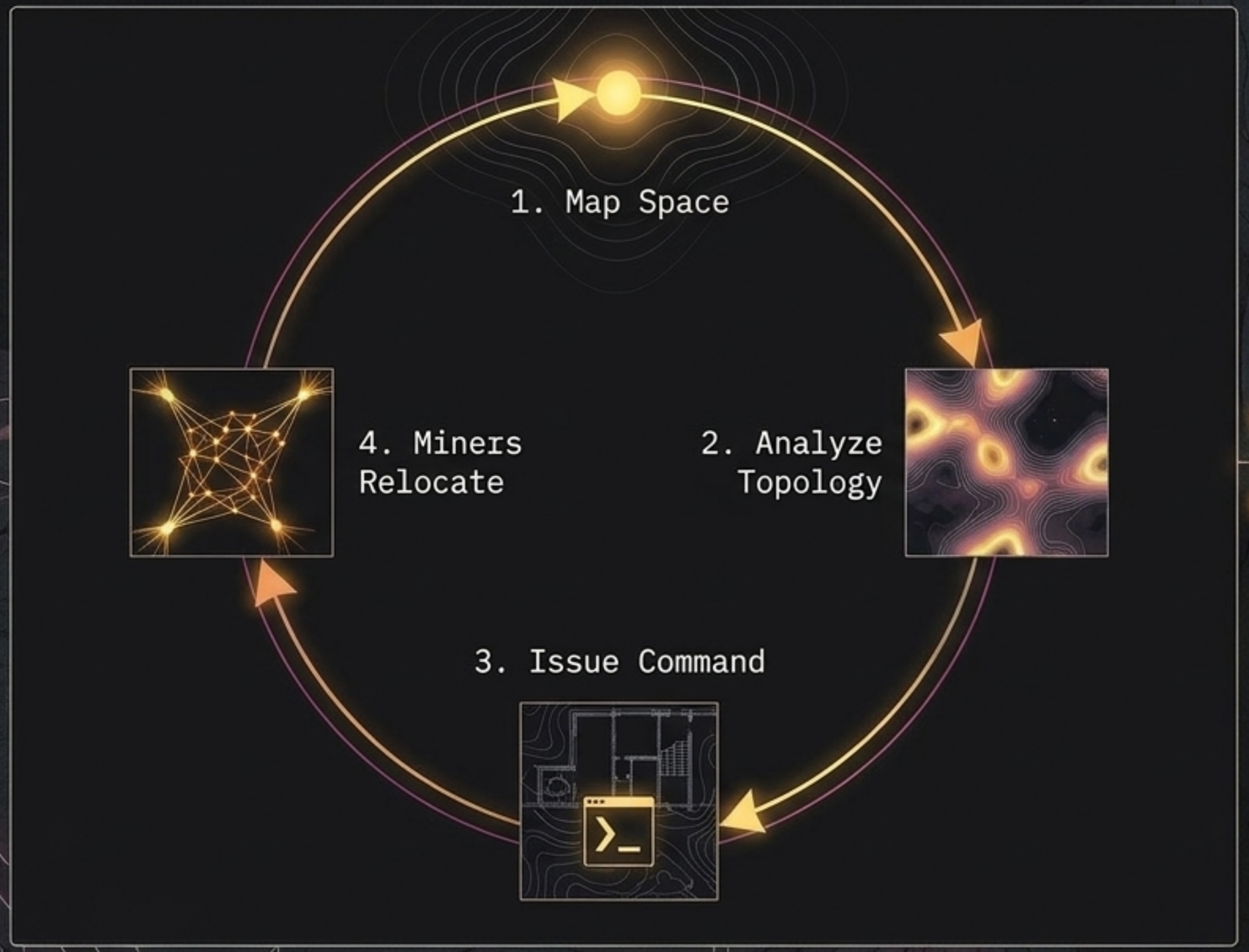
Mapper Graph



The nerve mapped over the novelty lens. Flaring nodes represent active exploration fronts.

Gemini Notebook

Closing the Loop



> Teleport

Instantly move a struggling branch directly to a newly discovered basin peak.

> Corroborate

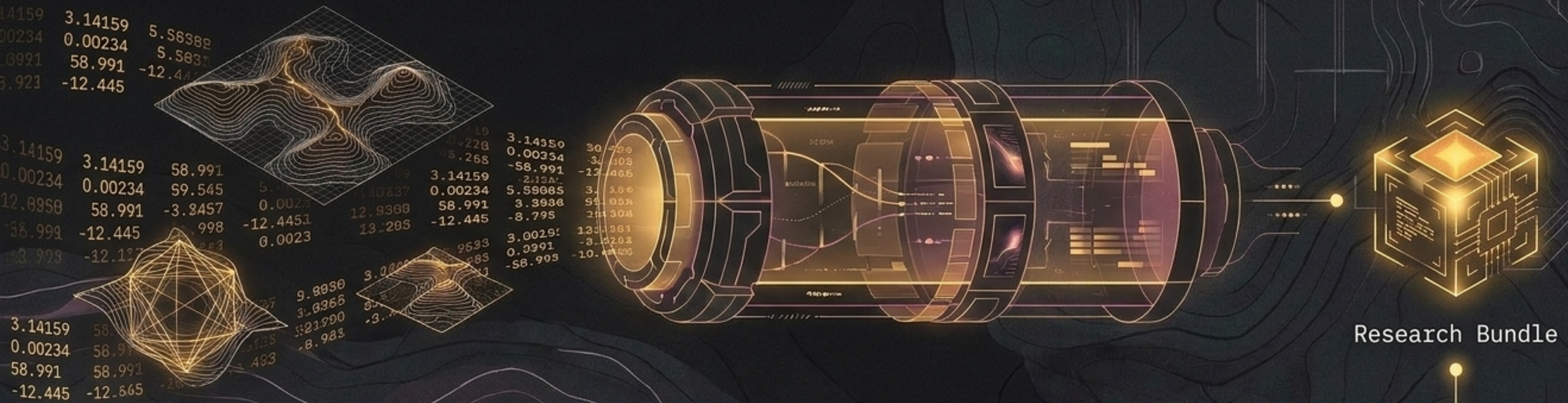
Target a novelty peak and attack it simultaneously from multiple spread persona lenses to test consistency.

> Probe-Next

Automatically command workers to target the sparsest topological gap or an unexplored void chamber.

Gemini Notebook

Automated Science & Export



Bundle JSON

Contains the manifest, full journal, execution states, and all 6 topological surfaces.

Data Arrays

Complete findings exported as flat, analysis-ready CSV files.

High-Res Imagery

3D Maps and exact shape graphs exported natively as PNG and Vector SVG.

Methods Generation

Auto-generates a pre-filled, mathematically precise methods paragraph ready for academic papers.

Gemini Notebook

Runnable Research Protocols

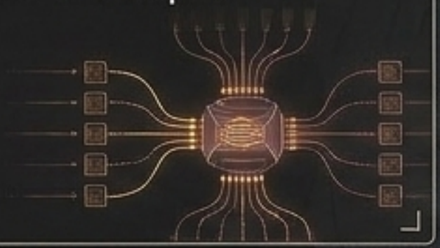
1. Knowledge-Frontier Cartography

Map the exact mathematical edges of what an LLM model knows.



2. Self-Consistency Corroboration

Spatially stress-test specific concepts across multiple AI personas.



3. Red-Team Strategy Archives

Spatially map adversarial prompts against model defensive boundaries.



4. Idea Discovery for Science

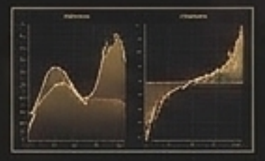
Find the unmapped topological voids



Find the unmapped topological voids between established academic disciplines.

5. Model Comparison

Run diff overlays to mathematically



Run diff overlays to mathematically compare two different models' latent spaces.

6. Novelty-Metric Research

Benchmark and test scoring metrics.



Benchmark and test KDE and topology scoring mechanisms in a live environment.

Gemini Notebook

Provenance & Reproducibility

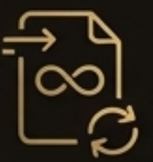
Self-Describing Journals

Every file permanently records model, endpoint, embed_model, vec_kind, and app version.





Vector Persistence



Vectors persist and reload. All shape and topological views remain fully interactive on archived runs forever.

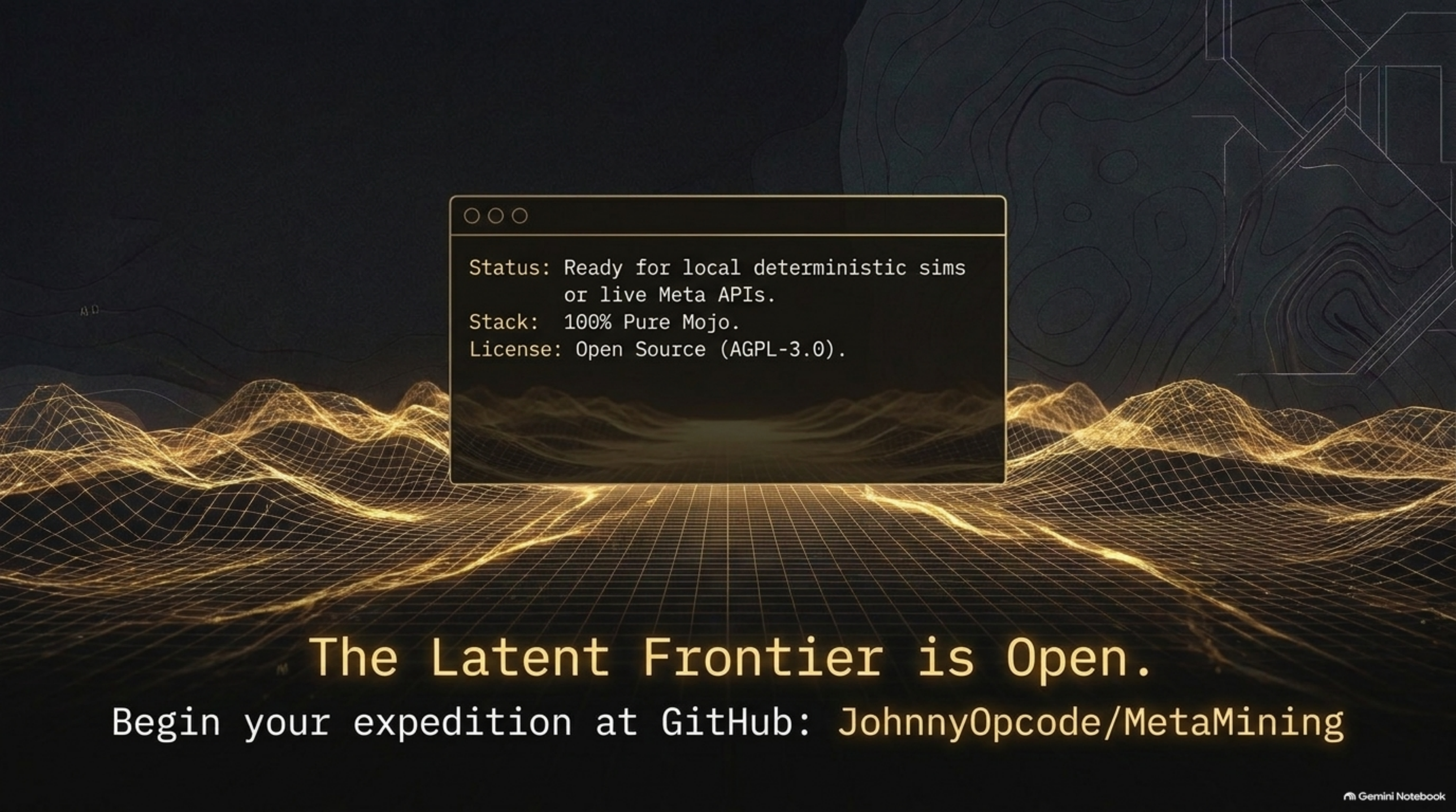
Legacy hash journals automatically recompute.

Shipped Benchmarks

Includes standard datasets providing highly reproducible reference runs for verifiable baseline testing.



Gemini Notebook



Gemini Notebook